

Hon Precision

#AI#Semiconductor#AdvancedPackaging

↔ Buy
α Portfolio

Price: TWD 5,800

Target Price: TWD 10,000

Upside: 72%

Total return: 75%

Risk: High

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Hon Precision

Bloomberg 7769 TT

RIC 7769.TW

Market cap (USDbn) 33.2

Average daily T/O (USD|m) 155

Hon Precision Key Reports

Initiation (2025/7/20) [Hon in on a home run](#)

Update (2025/7/31) [Hedged and lifted](#)

Update (2025/11/18) [Stronger for longer](#)

Update (2026/2/9) [Fast and Furious](#)

Update (2026/3/8) [Gaining Vision](#)

The 'Hon' Testing Velocity

We believe the current surge in the global semiconductor testing industry will not only persist but accelerate through CY28E, propelled by TSMC's planned quadrupling of advanced capacity from CY25–28E. Advantest's recent move to double its testing capacity by CY28/29E further underscores the industry's structural secular growth. This aggressive expansion is driven not only by rising demand for AI devices, but more critically by the rising complexity of chip design—marked by increasing chiplet counts, expanding reticle sizes, and TDP requirements. These trends are evident in new platforms such as Rubin Ultra, Feynman, MI450/550, TPU V9, and the new generation of agentic CPUs. Against this backdrop, we anticipate Hon will raise its capacity expansion to 40–50% CAGR in CY26–28E. We have raised our EPS estimates by 12-44% and our target price from NT\$6,200 to NT\$10,000. We now forecast earnings to grow 2.5x over the next three years. We maintain Hon in the Alpha Portfolio. Risks include market demand, execution and geopolitics.

More wafers, more testers, more handlers

TSMC is set to significantly accelerate capacity expansion across advance nodes (N3, N2, A16, A14) over the next two years, adding 70k/180k/170kwpm in CY26/CY27/CY28E to capture upcoming AI device opportunities. This will drive meaningful demand for WFE in CY27–28E, followed by a corresponding ramp in backend testing capacity demand in CY28–29E, typically a 9-month lag after WFE installation. In response to this secular demand growth, major ATE vendor Advantest has already announced plans to double its tester capacity to 10k units in CY28E vs ~5k in CY26E, while we suspect Teradyne will likely scale at a similar pace. Moreover, major OSATs (i.e., KYEC) also plan to expand testing capacity 2x, backed by active discussions from the leading foundry and IC design houses seeking to secure multi-year testing capacity through CY28E. The overall testing supply-demand environment is so severely constrained that major OSATs have recently raised hourly testing rates for the first time in history.

Rushing for faster capacity expansion to address underserved demand

Hon is a near-monopoly in the FT ATC handler market. Rising chiplet counts, expanding reticle sizes, and increasing power density are driving at least 30-40% higher tester/handler demand per AI device generation, with some platform requiring 100%+ more testing content vs predecessors. The increasing device complexity is driving structurally longer FT times by multiples across next-AI devices, including Mellanox networking ICs, WMCM, Vera CPUs, Humufish TPUs and ASICs. Overall, we expect FT handler demand to grow 80%+ in two years, implying a 40-50% CAGR. To address this demand, Hon has expanded its production footprint to five plants, and we expect another two plants to begin operations in 2H26 and CY28E—more than doubling its three-plant footprint last year and supporting a 40–50% capacity CAGR. This should help Hon address the currently underserved demand environment, where Hon has only been able to fulfil ~70% of clients' demand, while also alleviating potential competition arising from supply tightness.

KEY FINANCIAL AND VALUATION MATRIX

FYE Dec (TWD mn)	FY21A	FY22A	FY23A	FY24A	FY25A	FY26E	FY27E	FY28E
Revenue	12,292	13,469	9,489	13,992	30,271	54,132	91,819	130,986
OPN Profit	4,317	5,006	3,766	6,285	15,046	28,734	50,661	73,822
Net Profit	4,044	4,356	3,068	5,286	12,362	23,434	41,039	59,386
EPS (TWD), Diluted	67.3	27.1	18.8	32.7	75.5	129.9	227.6	329.3
EPS (TWD) consensus						110.8	167.9	294.3
PER (X)	86.2	214.2	308.2	177.2	76.8	44.6	25.5	17.6
PB (X)	35.4	80.0	76.8	68.2	16.4	14.6	10.3	7.0
EV/EBITDA (X)	53.6	162.3	243.2	141.7	58.4	33.8	18.8	12.5
ROE (%)	104.7	45.0	25.6	40.6	34.5	36.2	47.3	47.3
ROA (%)	60.3	29.2	18.4	27.3	26.0	28.9	38.6	39.8
Total return (%)	0.1	0.1	0.3	0.5	0.3	0.9	1.0	1.1

Source: Company, Bloomberg, Aletheia/TAG

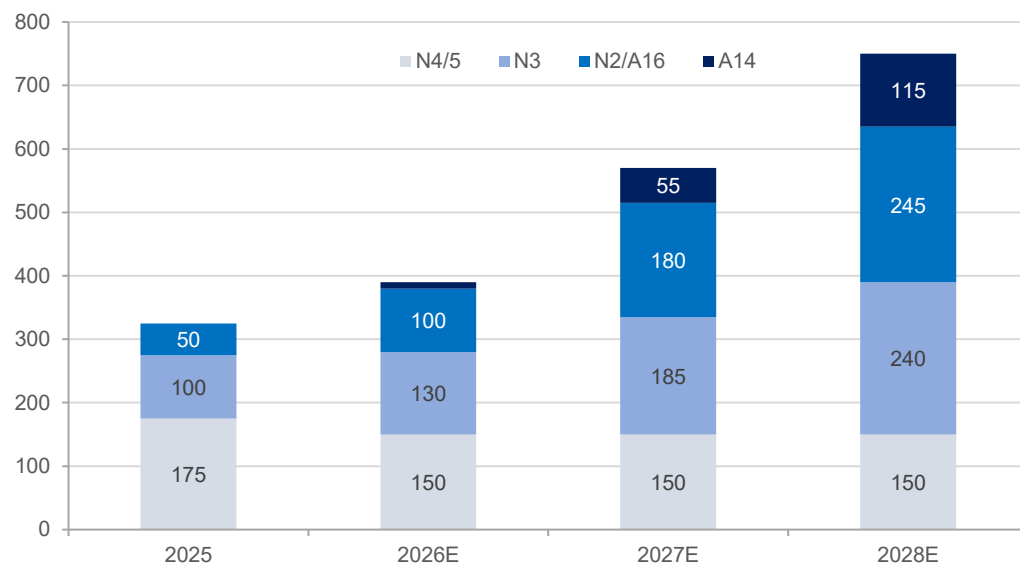
TSMC's monstrous multi-year capex driving secular testing cycles

TSMC has been increasingly aggressive about capacity expansion, including a pull-in of N3 in Japan, a potential earlier ramp of CoPoS, and an acceleration of SoIC expansion. We expect the new addition of TSMC's advanced nodes (<N5) to be 70k/180k/170k in CY26E /27E/28E, respectively. Total capacity for advance nodes will grow to 750k exiting CY28E, from 325k last year, 2.3x in three years.

Its back-end capacity expansion is even faster. We believe TSMC plans to grow its annual CoWoS wafer capacity by ~80%/50% to 1.15m/1.73m in CY26E/27E, followed by an earlier ramp of CoPoS by end-CY27E/1H28E; this is two to four quarters ahead of the original schedule. Some of its 8-inch fab will be repurposed for interposer manufacturing to accommodate the growing demand. SoIC capacity is also expected to grow to 20/40/80kwpm ending CY26E/27E/28E from <10kwpm now.

To fund such growth, TSMC could spend \$220-230bn capex in CY26E-28E, with two-thirds of this for WFE. This compares with \$101bn for the previous three years (CY23-25). If TSMC keeps its capex at its guided \$52-56b this year, the firm could spend \$80-90bn/yr in 2027/28E. We believe this aggressive expansion is driven by not only the existing AI GPUs, HBM base die, networking & ASICs but more recently, agentic AI devices such as CPU (x86 and Arm), DSP, LPU etc.

FIGURE 1: PROJECTIONS OF TSMC'S ADVANCED CAPACITY EXPANSION PLAN



Source: Aletheia/TAG

Against the backdrop of TSMC's aggressive front-end capacity expansion, with nearly 80% of incremental leading-edge capacity additions scheduled for CY27E/28E (Figure 2), we expect the WFE investment cycle to peak in CY27E and remain elevated through CY28E.

This should translate into a favorable backend equipment cycle, given the typical lag between front-end WFE investment and backend demand. We estimate it takes roughly three months for WFE installation, followed by another 5-6 months for wafer production ramp, implying a ~9-10 month lag from WFE procurement to backend testing demand.

As a result, we expect demand for testing equipment—including both testers and handlers—to accelerate meaningfully and peak no earlier than CY28E-29E.

FIGURE 2: PROJECTION OF TSMC'S ADVANCED CAPACITY EXPANSION PLAN IN TERMS OF PHASES

Node	Sites	Name	2025	2026	2027	2028
N3	TN	Fab18	-	-	P9+P10	P11+ (P12)
	Capacity		-	-	40	30
N2/A16	Sites	Name	2025	2026	2027	2028
	HS	Fab20	P1	P2	-	-
	KS	Fab22	P1	P2	P3+P4	P5
	TN	Fab18			P10	P11+ (P12)
	Capacity		50	50	65	55
A14	Sites	Name	2025	2026	2027	2028
	HS	Fab20	-	P3 (pilot)	P3	P4
	TC	Fab25	-	-	P1	P2
	Capacity			10	45	50
	Total capacity in TWN		50	60	150	135
N2/3/4 N6/7/3	Sites	Name	2025	2026	2027	2028
	Arizona	Fab21	P1AB	P2A	P2B	P3A
	Capacity		25	10	15	10
	Kumamoto	Fab23	-	-	P2A	P2B
	Capacity		-	-	15	25
	Total WW capacity		75	70	180	170

Note: Capacity figure is expressed in '000 wpm.

Source: Aletheia/TAG

In its FY1/26 earnings call, Advantest announced plans to further accelerate tester capacity expansion, targeting 10k units by March 2029 and ~5k units by March 2027, doubling tester capacity within two years. While Teradyne has yet to formally outline its expansion plan, we believe it will likely need to scale at a similar pace to keep up with industry demand, implying a potential global ATE capacity of ~14–15k units by end-2028, versus ~7–8k units currently.

Both ATE vendors have openly discussed their expectation that the wafer probe segment will outgrow final test over the next several years, leading final test's share of total ATE market to decline from 40%+ historically to 30–40% or lower going forward. We believe this trend is structurally driven by:

- rising chiplet counts, requiring more wafer probing stages across different die types; and
- increasingly complex advanced packaging flows introducing new intermediate test insertions between wafer probe and final test.

These new insertions include:

- partially assembled package testing (e.g., CoW testing prior to oS process),
- EIC/PIC bonded wafer testing before dicing, and
- die-level testing prior to final packaging.

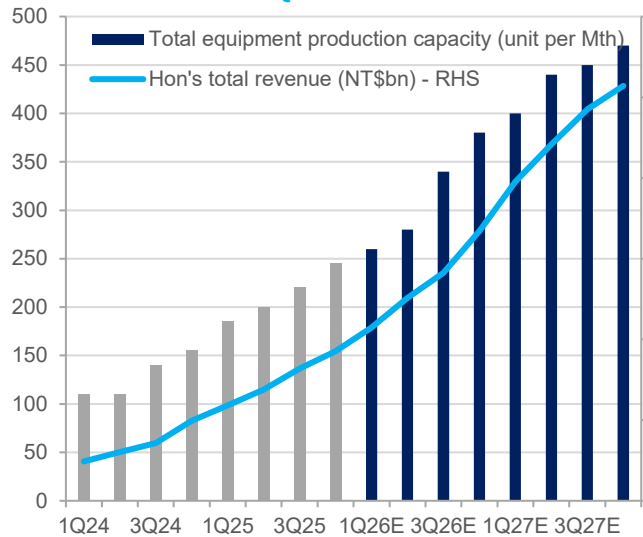
We categorize these as part of the broader wafer probe / shifting-left testing trend.

That said, we do not expect final test demand to decelerate—in fact, it should continue accelerating. Multi-die architecture introduces entirely new failure models absent in monolithic designs, including interconnect failures, TSV defects, thermal crosstalk, and known-good-die escapes. While some test-time reduction optimization exists at final test—

such as simultaneous testing of multiple functional blocks rather than serial scan patterns—similar optimization dynamics also apply to wafer probe, partially assembled, and die-level testing. Over time, some degree of cannibalization may emerge among these newer insertions as process flows mature.

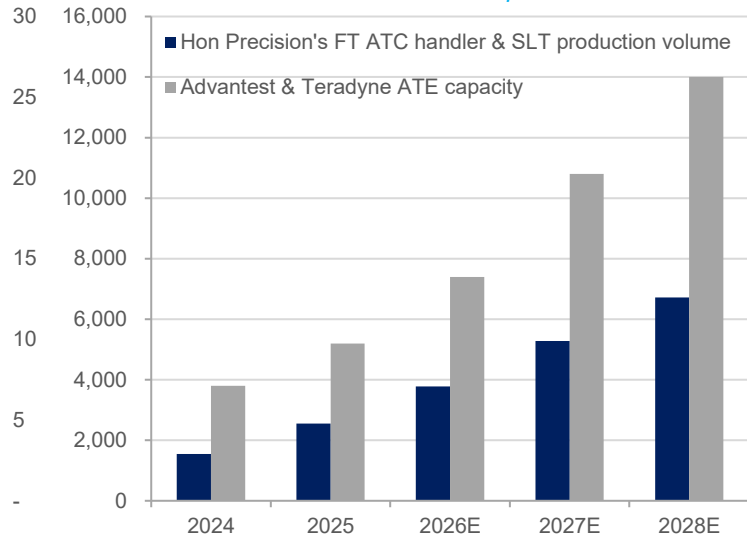
In our final test tester and handler market model, we assume final test accounts for ~40–45% of total ATE demand in CY25–26E, declining to ~35–40% in CY27–28E. However, despite the mix dilution, we still expect the final test segment to grow by 90%+ over the next two years, while the broader wafer probe ecosystem—including partially assembled and die-level testing—could more than double, supported by aggressive capacity expansion plans from both OSATs and ATE vendors.

FIGURE 3: HON PRECISION QUARTERLY CAPACITY GROWTH



Source: madpcb, Aletheia/TAG

FIGURE 4: GLOBAL ATE VS HON PRECISION FT/SLT CAPACITY

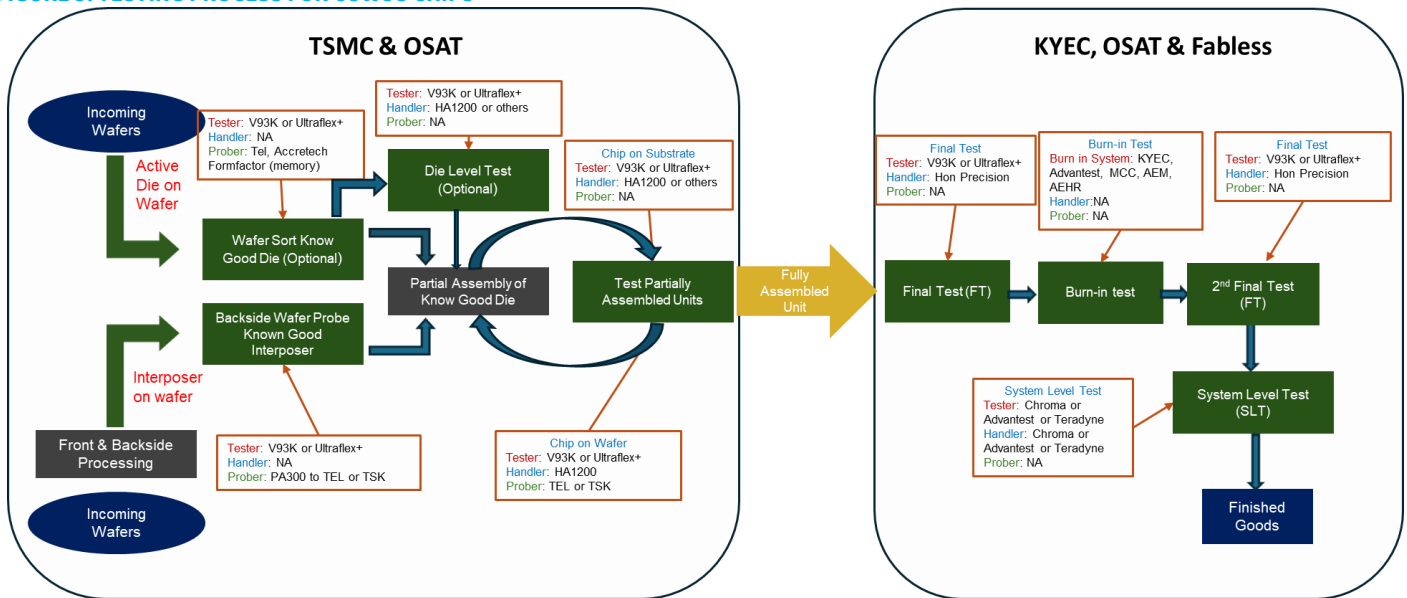


Source: MDPT, Aletheia/TAG

Complexity adds new insertions and extends CP/FT test times

Figure 5 below shows our current view of the testing process for CoWoS chips. The expansion of heterogeneous chiplet architecture used in CoWoS advanced packaging and the expanding reticle size of chips are driving more front-end testing to minimize scrap costs and mitigate overall system risks. For example, partial-assembly tests mentioned earlier effectively shift some test steps from final tests to CoW tests. That said, final test also grows meaningfully with packaged IC with growing chiplet tiles and reticle size.

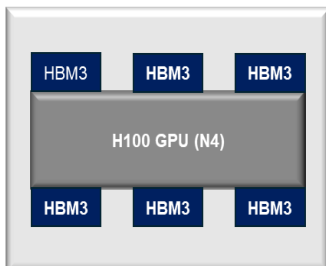
FIGURE 5: TESTING PROCESS FOR COWOS CHIPS



Note: *Here we focus more on Nvidia, but manufacturers can be interchangeable - this is only an estimated flow chart
Source: Aletheia/TAG

The following schematic diagrams represent the evolution of Nvidia AI chips, which have significantly increased in chiplets and reticle size.

FIGURE 6: NVIDIA HOPPER GPU



Note: Not to scale Source: Aletheia/TAG

FIGURE 7: NVIDIA BLACKWELL GPU (2024-26)

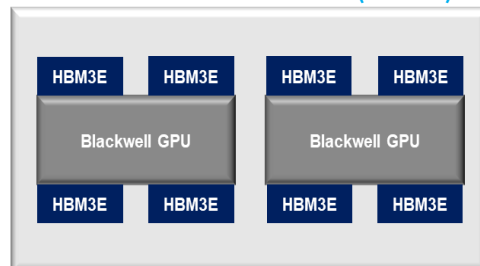


FIGURE 8: NVIDIA RUBIN GPU (2026/27)

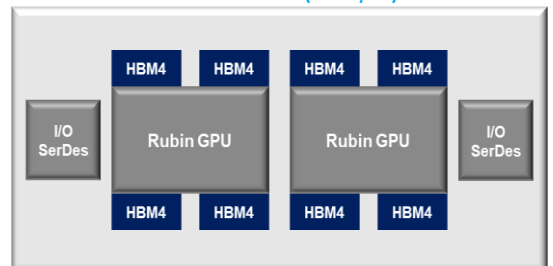
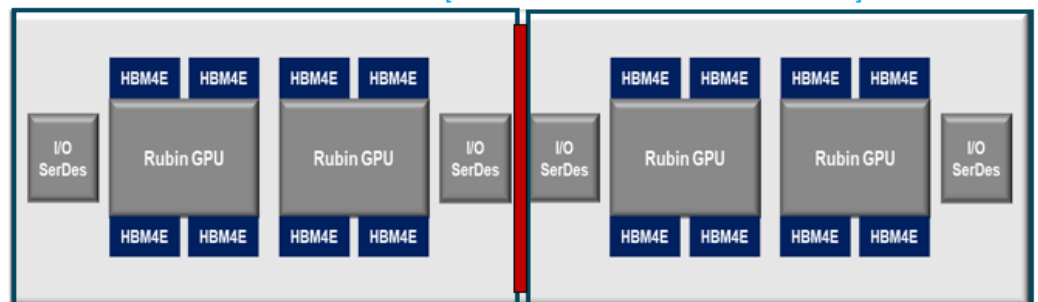
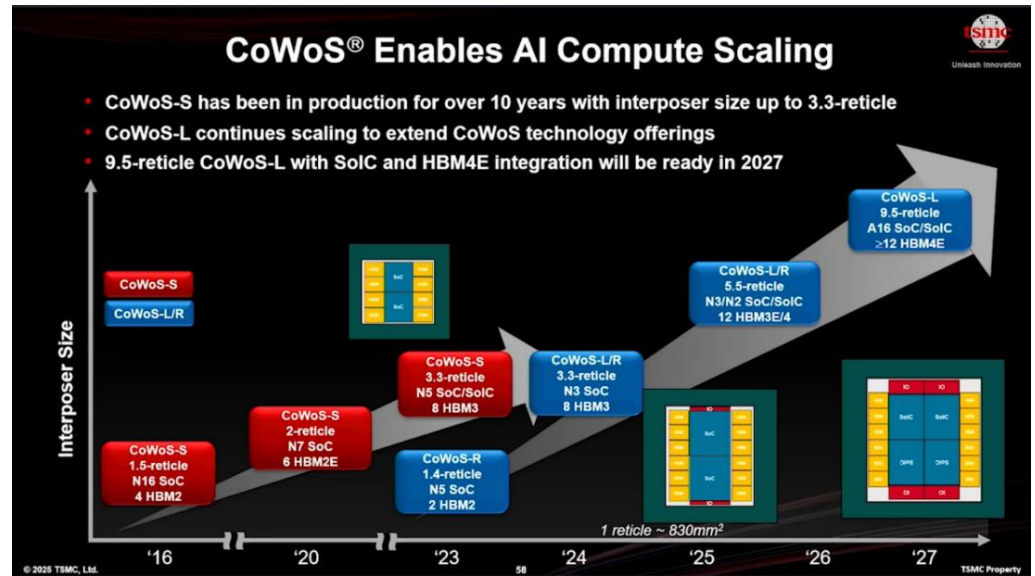


FIGURE 9: TOPOLOGY OF RUBIN ULTRA GPU [BLOCK IN RED REPRESENTS STICTCHING]



Source: Aletheia/TAG

FIGURE 10: TSMC'S COWOS PACKAGING ROADMAP

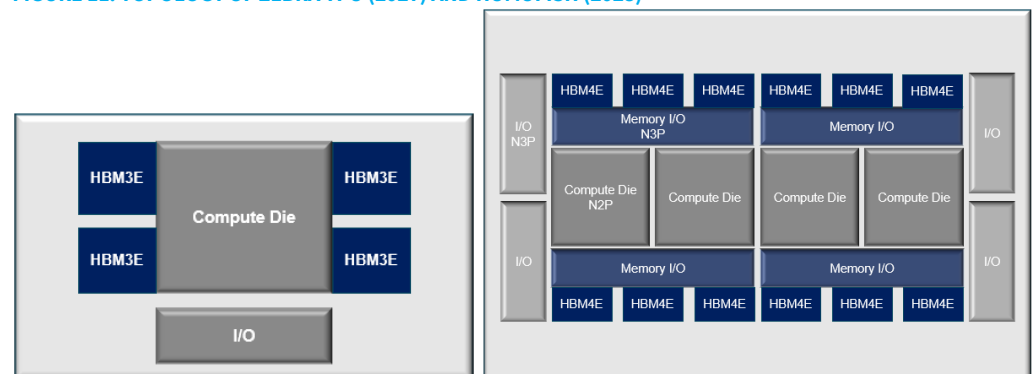


Source: TSMC

Increasing reticle size allows more chiplets to be attached on top of the interposer. Compared to Blackwell, Rubin separates the I/O functions from the GPU compute die to increase processing speeds. However, this introduces two newer types of wafers (one is the N3 wafer and the other is the N4 wafer) and therefore two additional wafer-sort testing processes, because both I/O dies have different designs using different process nodes. Therefore, compared to Blackwell, which only needs wafer sort for one type of Blackwell wafer, Rubin will need three different test stations with three different probe cards for three wafer types. While the two I/O dies are smaller than the Rubin GPU (therefore they will take less time), we estimate the total testing time spent on all three wafers to complete wafer sorts for Rubin could be 1.7-1.8x higher versus Blackwell. This means the demand for testing capacity would increase accordingly, if not more.

The increase in the types of wafers also increases for ASICs, as shown in the following figure, where we expect AWS, Google and Meta to all begin adopting three to four different types of wafers for their chiplet architecture starting in 2026-28. At the final test level, the FT time is highly related to reticle size and transistor density. We expect FT time to grow by 2x-10x for several upcoming AI devices versus predecessors, including Mellanox networking ICs, WCMC AP (vs InFO PoP AP), Vera CPUs (vs Grace CPU), Zebrafish TPU (vs Ghostfish TPU), Humufish TPU (likely close or surpassing Blackwell FT time), and others.

FIGURE 11: TOPOLOGY OF ZEBRA TPU (2027) AND HUMUFISH (2028)



Source: Aletheia/TAG

FIGURE 12: SPECIFICATIONS OF SELECTED AI GPU & ASIC DEVICES

nVidia	HVM	Architecture	Packaging	Logic core	Based die	I/O	Types of wafers
Hopper GPU	Mid-22	Monolithic	CoWoS-S	1x N4	-	-	1
Blackwell GPU	Mid-24	Dual-monolithic	CoWoS-L	2x N4P	-	-	1
Grace CPU	Mid-21	Monolithic	-	1x N4	-	-	1
Rubin GPU	Mid-26	Hete-Chiplet	CoWoS-L	2x N3	-	1x N3+1x N4	3
Vera CPU	Mid-26	Hete-Chiplet	CoWoS-R	1x N3	-	4x N3+1x N4	3
Feynman GPU	Mid-28	Hete-Chiplet	CoWoS-L+SoIC	4-8x A16	2-4x A16	4xN3	5

Google - TPU	HVM	Architecture	Packaging	Logic core	I/O	ASIC partners	Types of wafers
Trillium (V6)	4Q24	Hete-Chiplet	CoWoS-S	1xN5	1xN7	AVGO	2
GhostFish ² (V7p)	4Q25	Hete-Chiplet	CoWoS-S	2xN3E	1xN5	AVGO	2
SunFish (V8ax)	1H26	Hete-Chiplet	CoWoS-S	2xN3P	1xN4P	AVGO	2
ZebraFish (V8x)	4Q26	Hete-Chiplet	CoWoS-S	1xN3	1xN6	GOOG/MTK	2
HumuFish (V9x)	4Q27	Hete-Chiplet	CoWoS-L	4xN2	4xN3+4xN3	GOOG/MTK	3
TBD (V9ax) ¹	2027	Hete-Chiplet	CoWoS-L	N2	N3	AVGO	3

META	HVM	Architecture	Packaging	Logic core	Base die	I/O	SoIC	Types of wafers
MTIA 1/1.5	2023/24	Monolithic	-	1x N7	-	-	-	1
MTIA 2	2H24/25	Monolithic	-	1x N5	-	-	-	1
MTIA 3	2H26	Hete-Chiplet	CoWoS-S	1x N3	-	2x N4	-	2
MTIA 4 ¹	2027	Hete-Chiplet	NA	NA	NA	NA	NA	NA
MTIA 5	2028	Hete-Chiplet	CoWoS-L	16x N2	N2 or N3	2xN3	Yes	4

AWS	HVM	Architecture	Packaging	Logic core	I/O	SoIC	Types of wafers
Trainium 1	2023/24	Monolithic	-	1x N7	-	-	1
Trainium 2	2H24/25	Dual-monolithic	CoWoS-R	2x N5	-	-	1
Trainium 3	Mid-26	Dual-monolithic	CoWoS-R	2x N3	-	-	1
Trainium 4	2027	Hete-Chiplet	CoWoS-L	4x N2	1x N3	Yes	3

Note: 1-details to be determined or not available at the time of writing. 2-or known as Ironwood

Source: Aletheia/TAG (as of February 2026)

Agentic AI driving CPU testing demand

[this section is an excerpt from our recent AI Opportunities–CPU report, published on 29 March]

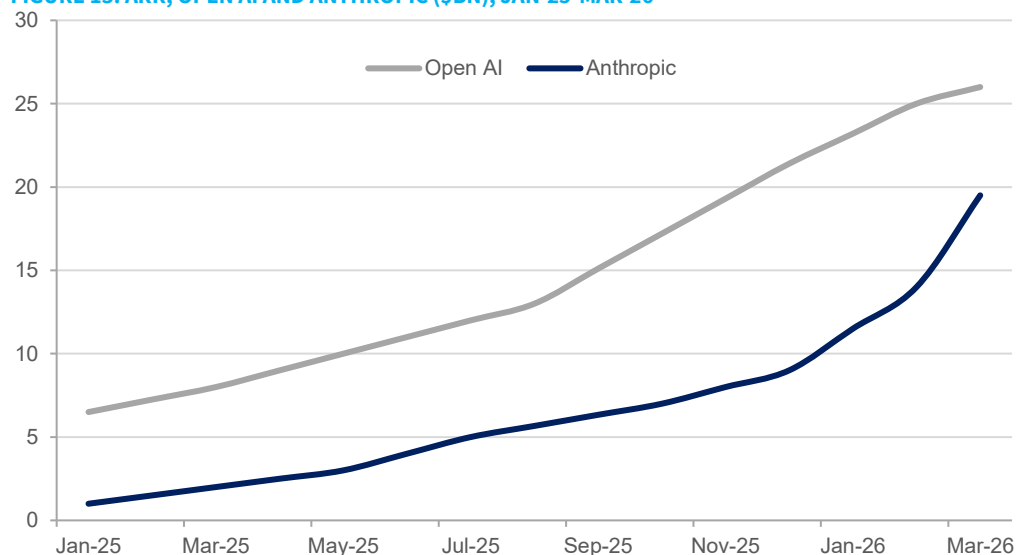
We estimate the total server CPU market amounted to ~25m units in 2025, with 80% of total being x86 based CPUs. We are anticipating ~20% YoY unit growth for this year to bring the market to 30m. The volume could be higher if there is no supply constraint. ARM-based CPUs, especially Google's Axion2 and NVDA's Grace/Vera, both have phenomenal growth, partially driven by the attachments with GPUs and TPUs. We expect AMD to grow its units by 30%+ YoY, driven by strong demand from its CSP customers. Intel, unfortunately, suffered from its capacity constraint and was down YoY in 1Q26E despite strong demand, but the company expects a comeback starting 2Q26E. Net-net, we estimate x86 will continue to have an 80% share of the units in 2026E.

We are just in the first year of agentic AI in 2026E. Compared with AI infrastructure investment in the past three years, the key difference now is that agentic AI can generate income. Anthropic is the best case. While its ARR growth in 2025 was already exponential from \$1b in January 2025 to \$9b exiting the year, it has further doubled in just three months to an estimated \$19-20b in March 2026, driven mainly by its Claude Opus 4.6. Open AI, which is more consumer-centric, was growing at a slower pace but has still expanded its ARR by threefold from <\$7b at the beginning of 2025 to \$21b exiting the year. In 1Q26, it has also grown its ARR by \$5b to \$26b now. Combined ARR growth was \$23b in 2025 and it has already reached \$15b in just one quarter this year. Their ARR should have grown even faster if there was enough compute power.

It is always difficult to pinpoint a CAGR for a longer-term timeframe, say three to five years. But there are several signs that make us believe that the unit growth for CPU units will be even stronger in 2027E than this year. We are modelling 30%+ YoY CPU unit growth in 2027E, with unit growth mainly coming from Nvidia, Axion (Google), and AMD. At its "ARM everywhere" event, ARM also guided for its data center CPU TAM to grow from \$50b in FY26E (ending March 2026) to \$100b+ in FY31E, or a doubling in five years. Our studies and conversation with other CPU makers suggest ARM may have overestimated its F26E TAM while underestimating F31E. In any case, all indicators are pointing to a stronger CAGR from here.

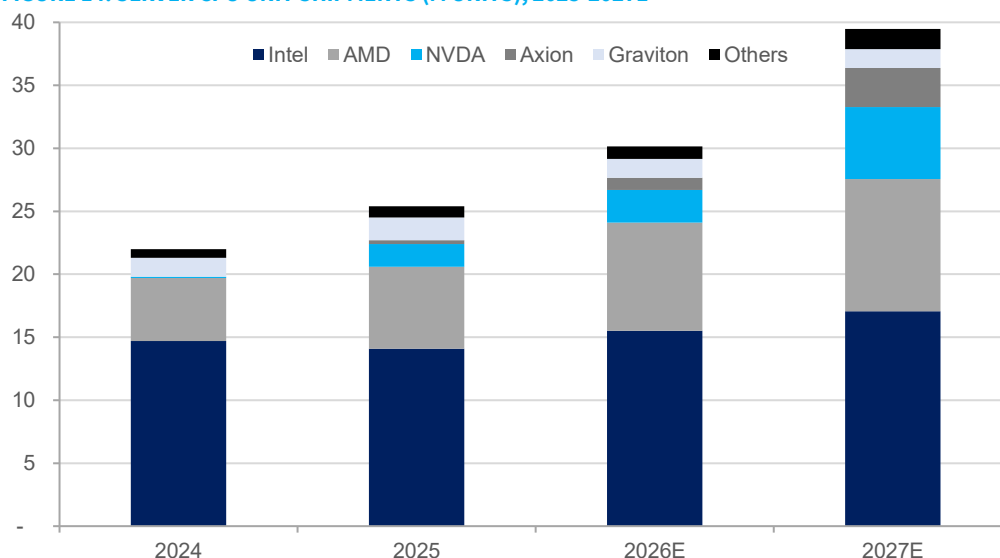
In the next few sections, we outline our key observations from the supply chain that drives server CPU unit growth.

FIGURE 13: ARR, OPEN AI AND ANTHROPIC (\$BN), JAN'25-MAR'26



Source: Aletheia/TAG

FIGURE 14: SERVER CPU UNIT SHIPMENTS (M UNITS), 2025-2027E



Source: Aletheia/TAG

Nvidia's SuperPod: GPU to CPU ratio could hit 2.4x vs 0.5x before!

Nvidia introduced its Vera Rubin SuperPod at its GTC 2026. Different from the previous generation Grace-Blackwell which had only one rack solution (NVL72), Nvidia released five different racks in the Vera Rubin generation, trying to address all different tiers of inferencing demand. In addition to the widely expected VR NVL72, which consists of Rubin GPU and Vera CPU, the other four racks include a Vera CPU rack, LPX rack (for LPUs), a Bluefield 4 STX storage rack, and a Spectrum switch networking rack.

This approach significantly increased the GPU-to-CPU attachment rate. Within this 40-rack SuperPod, the total Rubin GPU (16 racks) amounts to 1,152, while total Vera CPU numbers are as high as 1,088. The GPU-to-CPU ratio is 0.95, almost double from 0.5 for the GPU rack only. The story does not end here. There are 36x host CPUs in each LPX rack. In addition, Bluefield 4 shares the same logic die as its Grace CPU. Should we include all these CPU in our math, the GPU-to-CPU ratio will balloon to 2.4x, up from 0.75x in the original solutions (CPU plus Bluefield DPU in GB or VR rack).

We estimate Nvidia produced ~2m Grace CPUs last year and will produce 2.5-3m combined Grace and Vera CPUs this year. Looking into 2027E, we expect Nvidia to produce a total of 5-6m combined Grace and Vera CPUs, with Vera taking the majority. Grace and Vera CPUs in 2025/26E are mostly used for its Bianca and Strata module shipments, together with Blackwell and Rubin GPUs. There will be some standalone CPU shipments this year for in-house and external customer usage; [Meta is the first external customer](#). We anticipate there will be more CPU-only customers in the next year.

FIGURE 15: NVDA'S VERA RUBIN SUPERPOD: 7 DIFFERENT CHIPS AND 5 DIFFERENT RACKS



Source: NVDA, Aletheia/TAG

FIGURE 16: DETAILS OF NVIDIA'S NVL SUPERPOD STRUCTURE AND MEMORY CONTAIN

Devices	VR NVL72	Vera CPU	Groq LPX	BF4 STX (storage)	Spectrum-6 SPX (networking)
GPU	72x	-	-	-	-
CPU	36x	256x	32x	-	-
BF4 DSP	18x	64x	32x	72x	-
LPU	-	-	256x	-	-
NVLink	18x	-	-	-	-
SP6	-	-	-	-	16x
# of racks per Superpod	16	2	10	8	4

Source: NVDA, Aletheia/TAG

AWS: new inferencing server form factors

Nvidia is not alone. We also learned from the supply chain that AWS is requesting some new CPU server form factors. For instance, the first model will be six Intel CPUs (Granite Rapids) sitting on the same server board, with each of them equipped with six R-DIMM and each DIMM will support 64GB server DRAM. This compares with the usual CPU server board of one or two CPUs per board. DRAM density will also be significant per board at 2.3TB. More importantly, we believe this is not the only SKU and there will be more to follow-up.

Notably, AWS has developed its in-house ARM-based CPU (Graviton) since 2018, and it is already the fifth generation this year. Its Graviton 5 will be fabricated at TSMC's N3. We expect AWS to continue to use its in-house CPU for ~50% of its demand, while it will also continue to source x86 CPUs aggressively, evidenced by the inferencing server mentioned above.

Google: strong growth for both in-house and x86 CPUs

Google launched its Axion CPU in late 2024, but we expect volumes will only take off from its second-generation Axion 2 this year. Nevertheless, the driver for Axion2 this year is mainly to ship together with its TPUs. Its current TPUv7 (Ironwood, Ghostfish) is shipping together with Intel's CPU, and the TPU-to-CPU ratio is 2:1. However, we believe its TPUv8 families, including Sunfish from Avago and Zebrafish from MediaTek, will be equipped with Axion2. We forecast combined Sunfish and Zebrafish shipments will reach 1.3m+ this year, suggesting ~700k demand for TPU racks. Total Axion shipments should reach 1m+ this year accordingly. Looking into 2027E, we are forecasting total TPU shipments of 6m+, with 5m+ from TPU v8 and v9 families. This implies demand of 2.5m+ Axion CPUs for TPU racks, and 3m+ total shipments should not be difficult.

Google is also buying external x86 CPUs aggressively. For instance, we believe it requested up to 3-4m CPUs from AMD earlier this year, compared with ~1m in 2025E. There is no way that AMD can fulfill such demand given the current supply constraint. We believe the demand strength will continue into 2027E. Most of Axion unit growth is captured by its TPU racks and Google will continue to need x86 CPUs for its agent and other workloads.

AMD: strong order backlog accelerates since 3Q25

AMD was already growing its server CPU shipments from a 1.2-1.4m/qtr run rate in 2024 to around 1.5m/qtr in 1H25, thanks to its Bergamo and Turin CPUs which are optimized for CSPs. The run rate further accelerated to 1.7-1.9m/qtr in 2H25, and we believe it has jumped to 2m/qtr+ this year. Meta was the first client to top up its demand for AMD back in late 3Q25, soon followed by MSFT. As aforementioned, Google has also come in to request multiple times volume in 2026E vs 2025E, but AMD already cannot fulfill this demand.

We believe AMD is aggressively trying to secure capacity for the years to come. AMD CEO Lisa Su was widely reported to have visited Korea and Taiwan recently to secure memory and foundry capacity. In addition, AMD is aggressively requesting CoWoS-like capacity from OSATs, which should be related to its upcoming Venice CPU. ASE just announced its subsidiary SPIL bought an old fab from Innolux for TWD6.3b (\$200m). We believe this is related to its LEAP business, for which CoWoS-like packaging (FOCUS and FOCUS-Bridge) will become important in 2027/28E. If everything goes smoothly, this new fab should be able to serve customers and contribute to ASE revenues before mid-2027E.

Peer comparison: AMD vs ARM-based CPUs

ARM-based CPUs can offer as much as 50–60% better power efficiency compared to x86 CPUs; however, AMD's processors maintain a strong value proposition in the world of Agentic AI. Below, we provide a detailed specification comparison between AMD's Turin-X/Venice-X and four major ARM-based competitors: AWS Graviton 5, Google Axion 2, NVIDIA Vera, and the upcoming Arm AGI CPU. In brief, while ARM-based peers excel in energy density and raw memory throughput, AMD's Turin-X provides larger local data residency, which results in lower operational latency. Furthermore, its software compatibility remains superior to its ARM-based counterparts.

Architecture: Cache Residency vs. Memory Bandwidth

The Graviton 5, Arm AGI, and NVIDIA Vera CPUs are equipped with wider memory bandwidth compared to Turin-X. The former two support DDR5-8800 (delivering up to 140.8 GB/s), while Vera utilizes an LPDDR5X interface on SOCAMM2, reaching up to 1.2 TB/s. Turin-X operates on DDR5-6400 with a maximum theoretical bandwidth of 102.4 GB/s.

Nevertheless, Turin-X features the largest on-die memory in its class, with over 1.2 GB of SRAM density (L2+L3 cache)—roughly 2–4x that of its ARM-based peers. In Agentic AI, Time to First Token (TTFT) and complex task orchestration are critical metrics. AMD's "Cache-First" architectural choice lowers operational latency by keeping the "working set" of an agentic loop—including scratchpads, tool-call metadata, and small-model parameters—entirely within the SRAM. By doing so, AMD avoids the power-hungry and high-latency penalties associated with external memory fetches, effectively offsetting the power efficiency benefits of its ARM competitors.

Software Compatibility and Ecosystem Reliability

A critical Total Cost of Ownership (TCO) factor often overlooked is Software Portability. Turin-X runs on the x86 architecture, which remains the native target for the vast majority of enterprise software and Python-based agent frameworks. Migrating to custom ARM silicon, such as Axion 2 or Graviton 5, often requires significant engineering resources for optimization and debugging. While it is easier for Cloud Service Providers (CSPs) to run internal workloads on their own customized ARM CPUs, their external customers—whose platforms are typically built on x86—face a much steeper transition.

The Future: Venice-X

The upcoming Venice-X is designed to neutralize ARM's current bandwidth advantage. Venice-X is expected to utilize 16-channel MCR-DIMM technology, which will provide the CPU with up to 1.6 TB/s of bandwidth, effectively surpassing the throughput of NVIDIA Vera. Additionally, it will expand on-die SRAM density to an estimated 2.3 GB+. This makes Venice-X potentially the most balanced engine for Agentic AI, combining the world's highest cache-hit rates with elite-tier memory throughput.

FIGURE 17: SPECIFICATION OF SERVER CPU

Supplier	AMD	AMD	Google	AWS	ARM	Nvidia
Product	Turin-X	Venice-X	Axion2	Graviton5	AGI CPU	Vera
Architecture	Zen 5 (x86)	Zen 6 (x86)	Neoverse N3 (ARM)	Neoverse V3 (ARM)	Neoverse V3 (ARM)	Olympus (ARM)
Total SRAM density	1,636MB	~2,800GB	~200 MB	~400 MB	640 MB	338 MB
Max Core Count	128 cores	256 Cores	72 vCPUs	192 Cores	136 Cores	88 Cores
Peak Clock	5.0 GHz	>5.0 GHz	~3.5 GHz	~3.7 GHz	3.7 GHz	~3.5 GHz
DRAM Standard	12-Ch DDR5-6400	16-Ch MCR-DDR6	DDR5-5600	DDR5-8800	12-Ch DDR5-8800	8x SOCAMM LPDDR5X
Max DRAM Density	6 TB	12 TB	768 GB*	1.5 TB	6 TB	1.5 TB*
TDP	400W-500W	500W-700W+	200W-300W	250W-350W	300W	150W-250W
TSMC process node	N3/N4	N2	N3	N3	N3	N3

Note: * Axion2 C4A variant carries up to 768GB memory density, up from 192GB in Tau T2A – the first generation of Axion CPU. Vera can support up to 1.5TB of LPDDR5X density, up from 520MB in Grace.

Sources: Companies, Aletheia/TAG

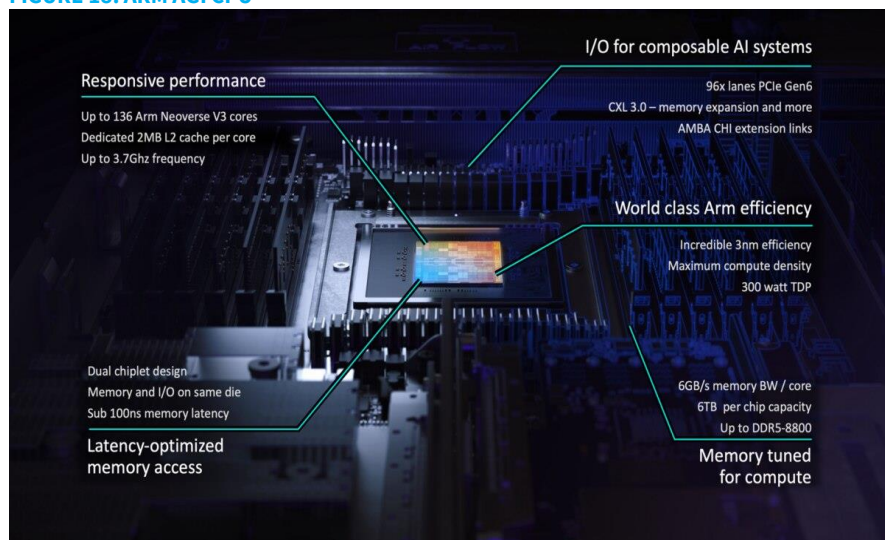
ARM: launching AGI CPU

On March 24, 2026, ARM launched its first silicon product named Arm AGI CPU, an ARM-designed CPU for AI data centers, built to address a rising class of Agentic AI workloads. According to ARM, agentic AI systems are the primary catalyst for its foray into silicon solutions as these systems demand CPU become the “pacing element” in AI infrastructure for orchestration, scheduling workloads, managing memory and coordinating multi-agent execution flows which is very much in line with our thesis. ARM explicitly stated that ecosystem partners demanded deeper solutions beyond IP licensing. To address this, ARM extended its platform from CPU IP to Compute Subsystems (CSS) to now full silicon products.

The Arm AGI CPU is based on the Arm Neoverse V3 platform and is optimized for rack-scale AI infrastructure. The Arm AGI CPU is built on the TSMC N3. It has up to 136 Neoverse V3 cores running at 3.2 GHz all-core and 3.7 GHz boost across two dies with 300W TDP. It supports 2x 128-bit SVE (Scalable Vector Extension) and has an L2 cache of 2MB per core. The device also supports 12 channels of DDR5 memory up to 8800 MT/s, which translates to memory bandwidth of ~6 GB/s per core. The I/O includes 96 lanes of PCIe Gen6 and CXL 3.0 for memory pooling and expansion. It also supports AMBA CHI links for accelerator integration.

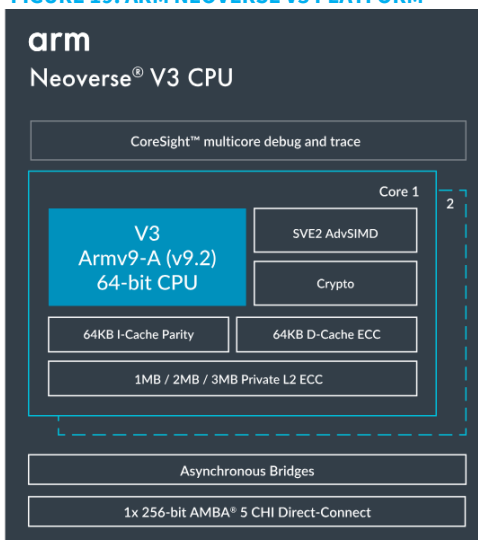
ARM referenced its new Arm AGI CPU 10U dual node server platform, which adopts dual-node (2N) configuration, enabling two independent compute nodes within a single 10U chassis. This design allows data center operators to double the nodes per rack unit, increasing overall compute density in the same physical footprint. As two AGI CPUs fit per blade, a standard rack can hold 30 blades, meaning 60 AGI CPUs resulting in 8,160 cores in total. Whereas, a liquid-cooled 200kW rack configuration can house 336 chips, resulting in 45,696 cores in total. ARM has partnered with Supermicro to build liquid cooled configurations. ARM claims the AGI CPU delivers >2x performance per rack compared to x86 systems achieved through higher memory bandwidth, better per-thread performance and higher usable thread count.

FIGURE 18: ARM AGI CPU



Source: Arm

FIGURE 19: ARM NEOVERSE V3 PLATFORM



Source: Arm

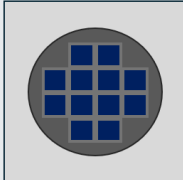
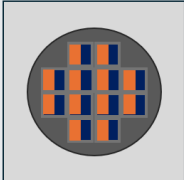
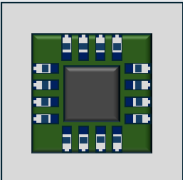
Implications to Hon Precision

Hon Precision is AMD's primary FT ATC handler partner that installs Advantest's ATE, particularly for their HPC and CPU chips. We believe the majority of Hon's handler system sales is sold to Tongfu in Malaysia, which is the main testing OSAT for AMD. On the other hand, AMD also has outsourced projects to ASE/SPIL in Taiwan. Nvidia's Vera is also currently only using Hon's handlers, where we see new orders entering Korea's Amkor facility. We believe the Graviton CPU developed by Annapurna labs also heavily utilizes Hon's ATC handlers. In addition, Google's Axion relies exclusively on Hon's handler and compatible with ATEs that potentially would be a dual source of both Advantest and Teradyne. Based on these new CPU projects, we expect the strong growth in CPUs to benefit Hon.

CPO test insertions

CPO testing involves four testing insertions given the integration of electrical and optical components and their interconnections with networking switches. While the high-volume ramp of CPO testing equipment may commence in 2H26-2027, all potential vendors are optimizing their test systems and TSMC and OSATs still evaluate multiple solutions.

FIGURE 20: CPO TEST INSERTIONS

CPO Test Insertions			
Insertion 1	Insertion 2	Insertion 3	Insertion 4
			
Photonic IC (PIC) (modulator, power detector)	Electric IC On PIC wafer (driver, TIA)	Optical Engine (fiber attach, DSP, interposer)	ASIC with CPO (packaged module)
E & O DC. (power, loss, dark current)	E/O, O/E, O/O, high-speed, RF/S-parameters	Full calibration/DC, high speed functional, optical lookback: RF/S-parameters	Full system functional validation, optical lookback

Source: Aletheia/TAG

The figure above describes the test process flow for CPO across the manufacturing flow. In CPO systems (for example, switch IC + optical engines integrated in a package or on the same application board), test coverage must move from wafer-level photonics IC (PIC) & Electrical IC (EIC) → electro-optical integration → optical engine → final ASIC package validation.

Each testing insertion is a test point in the manufacturing flow where yield learning and functional validation occur before progressing to the next integration stage:

Insertion 1: PIC/EIC wafer-level test

- **Photonic IC (PIC)**
 - PIC, normally made on the 65nm base die dimensions or 28nm PIC process, contains (1) fundamental passive and (2) active optical components, such as silicon MRM or MZM modulators (which encode electrical data onto light), waveguides (which route the light), and power detectors or photodiodes (which convert received light back into electrical currents).
 - The testing focus here is strictly on Electrical and Optical direct current (DC) tests, where automated optical wafer probers align optical fibers to the grating couplers on the wafer to measure the insertion loss (how much light is lost travelling through the waveguides and components).
 - A “Dark current” test is the key measure of the leakage current in the power detectors (photodiodes) when no light is present. High dark current indicates manufacturing defects and will lead to a noisy receiver, causing data errors.
- **Electrical IC (EIC)**
 - EICs, often fabricated on the N7 foundry process, often contain the TIAs (transimpedance amplifier) for the receiver side and the drivers for the modular on the transmitter side.

- The EIC wafer testing is more conventional than the PIC wafer probe. The testing focus is leakage current, power consumption, SerDes signal integrity, and some high-speed functional testing. The difficulty normally lies in extremely high frequency signals (that is, a 224G PAM4 may require probe cards with very-low-loss RF transmission), probe contact pitch of <40um, and thermal effects.
- **Equipment vendors for Insertion 1**
 - **(1) Engineering stage:** Keysight's tester is installed with FormFactor's CM300xi prober for both PIC/EIC wafer probes. No probe card is involved in the engineering stage.
 - **(2) MP stage:**
 - **EIC testing:** Our check suggests the POR vendors are Teradyne on ATEs, Tokyo Electron (TEL) or Tokyo Seimitsu (TSK) on probers.
 - **PIC testing:** (a) **Advantest camp:** Advantest (ATE) works with FormFactor (optical test module) and TEL (prober) on a TRITON PIC solution; (b) **Teradyne camp:** Teradyne (ATE) works on a PIC solution using its subsidiary, Quantifi Photonics, for optical test module, potentially also using Formfactor's CM300xi. We think FormFactor and Advantest's solution may be more mature during the initial valuation phases; and (c) **Chroma's** Model 586XX provides PIC reliability and wafer acceptance test.

Insertion 2: EIC+PIC (COUPE) wafer integration test

- This stage represents a critical heterogeneous integration using advanced packaging architecture like TSMC's COUPE (Compact Universal Photonic Engine) where EICs are hybrid bonded onto the PIC wafer.
- The testing focus here is the electro-optic conversion interfaces. Key measures include **(1) E/O (electrical-to-optical):** injecting an electrical signal into the EIC driver and measuring the optical output from the PIC modulator to ensure the transmitter path functions; **(2) O/E (optical-to-electrical):** injecting an optical signal into the PIC photodiode and measuring the amplified electrical output from the EIC's TIA to verify the receiver path; and **(3) high-speed and S-parameters:** sweeping the RF to measure the bandwidth, signal integrity, and impedance matching between the EIC and PIC.
- There are also two different lists of equipment vendor for the EIC/PIC bonding test
 - **(1) Engineering stage:** Both Keysight and Anritsu testers are installed along with MPI's double-sided wafer-level prober for the EIC/PIC bonded wafer. No probe card is involved in the engineering stage.
 - **(2) MP stage**
 - There are also several camps for the EIC/PIC bonded wafer test. As yet, **Teradyne camp** is the only testing solution that has passed TSMC's MP qualification, including Teradyne's ATE and Ficontec's WLT-D2 system. However, we learned that Ficontec's double-sided wafer-level prober comes in at only single-site. Nvidia has approached a Taiwanese backend company to work with Ficontec in optimizing its WLT system.
 - Another alternative supply chain is **Advantest camp**—Advantest provides ATE, MPI provides double-sided die level probers, and PI

or Keysight provides optical test module. Our check suggests that Advantest's solution is now engaging with TSMC for engineering and, thanks to MPI's double-sided wafer level prober yield superior throughput and signal integrity metrics, it currently received positive feedback at engineering. Just this month, MPI has just completed the MP version wafer level prober and is set to send for TSMC's MP qualification soon.

- Beyond the two ATE camps, **Chroma** noted that they have received EPIC insertion 2 testing equipment orders.

Insertion 3: EIC+PIC (COUPE) die level test

- After the EIC-PIC engine is diced to make an EPIC optical engine (OE), it will be attached to a silicon carrier, which will be attached with a receptable that will later be inserted with FAU (fiber array unit). The FAU will be either attached through grating coupling or edge coupling. After the FAU/receptable is assembled, the external laser source (ELS) will be plugged with FAU to connect to the OE die to send through the CW laser light source and perform die-level testing.
- The components that need to undergo testing at Insertion 3 include: (1) the CW laser ELS utilizing Chroma's photonic tester; (2) FAU/receptable that relies on active alignment assembly (UV) tools and inspection systems from the likes of Allring Tech or ADST; and (3) the OE assembled with receptable/FAU and connected with ELS may rely on testing vendors such as Advantest camp, Teradyne camp, and Chroma.
- For the OE die test, two main testing methodologies are being evaluated at the moment: a final test (FT)-like approach using a socket and load board, and a chip probing (CP)-like approach using probe cards. We noticed that for the current generation CPO device, the FT-like approach is preferred but the CP-like approach may be considered for the upcoming CY27-28E CPO devices.
 - The FT-like approach offers advantages in high-current testing capability and the ability to execute a broader set of test items and patterns. However, its alignment precision may be limited, as FT socket pitch typically ranges around 200–300µm.
 - In contrast, the CP-like approach enables much finer pitch alignment (<50µm) using probe cards, making it more suitable for fine-pitch optical/electrical interfaces. The trade-off is lower current capability and more limited test pattern coverage compared with FT-based testing.
- Several equipment vendors address the CPO insertion 3:
 - (1) **Chroma** noted that it has received purchase orders for insertion 3. Our check suggests that, currently, Chroma is the only POR for this insertion for its Model 58721 that will be installed at SPIL site;
 - (2) **Teradyne camp:** Ficontec also offers a singulated double-sided CPO tester (DLT-D1) for this insertion, supporting up to three parallel test heads. Initially, Teradyne had been working with Ficontec for the die-level prober solution; however, our checks indicate that Teradyne has since moved toward an alternative prober supplier, following difficulties in achieving the required prober quality and performance.
 - (3) **Advantest camp:** MPI also has its die-level CPO probers working with Advantest (ATE) and a Germany optical module company. Besides, SemiProbe, Micromanipulator and WinWay all offer double-sided probing systems or load board/sockets.

Insertion 4

- The heavily vetted optical engines from insertion 3 are surface-mounted onto a substrate alongside a central host ASIC or a high-bandwidth network switch.
- The testing focus is full system-level integration and functional validation. Key measurements include full system functional validation that the entire module will be powered up to ensure the host ASIC can communicate successfully through the package substrate, into the optical engines, and out through the optical fibers. This tests for: power delivery stability, thermal management under full load, and crosstalk between the optical engines and the ASIC. Another key measurement is optical lookback that performs final loopback tests, but this time driven entirely by the host ASIC's internal logic, to verify the ultimate system-level bit error rate.
- Mellanox has been working with SPIL on a manual PCB level test handler for insertion 4 development. However, it will ultimately need to scale via automation production and testing lines. For Nvidia's CPO devices, **Hon Precision** (handler) is working with **Teradyne** (ATE) and **Chroma** are also bringing out their own test handler solutions.

Hon's opportunities in CPO

Hon is currently engaged with Nvidia across three CPO switch programs. The company has already shipped FT handlers for electrical testing of ASICs, OEs, and integrated ASIC+OE modules.

Notably, Hon is the final test handler provider for test insertion #4, including BER testing and optical-electrical co-testing of integrated ASIC+OE modules. The development is targeted for completion in late-3Q26 or 4Q26 and initial deliveries expected in 2027. In parallel, Hon is also developing an SLT handler for OE optical testing.

Beyond this key customer's multiple CPO switch projects, Hon is also actively engaging with other ASIC customers on their respective CPO programs, further expanding its potential exposure to the emerging CPO testing ecosystem.

Our initial gauge for Hon Precision's CPO TAM for Nvidia's upcoming optical-electrical co-test of integrated ASIC+OE modules in CY27E is around \$100mn+ assuming the final test tester and handler demand at insertion #4 to be the same as insertion #3 die-level test, which is likely to see at least 100-unit demand.

Earnings estimate change

We raise our CY26/27/28E EPS estimates by 12%/30%/44% to NT\$130/NT\$228/NT\$329, respectively, to reflect our higher production capacity assumptions, now projected to grow by approximately 50%/40%/30% during CY26E–28E.

Hon Precision has already leased and acquired two additional plants beyond its original three-plant footprint, and we suspect the company may add another two plants over the next two years, potentially bringing the total to seven plants.

We expect blended equipment ASPs to grow by ~15% p.a. over the next two years, driven by the increasing mix of AOI-integrated handler systems—which carry 10–15% higher ASPs—rising from 10–20% of mix in CY26E to 50–80% over the following two years. Additional upside should come from the replacement cycle toward higher-grade FT ATC and SLT handler systems, particularly for 8–9× reticle-size AI devices and >3–5kW TDP platforms.

FIGURE 21: HON PRECISION - ESTIMATE CHANGE

(TWD mn)	FY1Q26E			FY2Q26E			FY26E			FY27E			FY28E		
	New	Prev.	Var (%)	New	Prev.	Var (%)	New	Prev	Var (%)	New	Prev	Var (%)	New	Prev	Var (%)
Sales	10,725	9,894	8.4	12,580	11,560	8.8	54,132	47,473	14.0	91,819	70,032	31.1	130,986	90,465	44.8
Gross profit	6,167	5,986	3.0	7,296	6,959	4.8	31,651	28,951	9.3	54,968	43,259	27.1	79,720	56,183	41.9
Operating profit	5,523	5,392	2.4	6,592	6,312	4.4	28,734	26,384	8.9	50,661	39,887	27.0	73,822	52,020	41.9
Net profit	4,529	4,310	5.1	5,405	5,058	6.9	23,434	21,041	11.4	41,039	31,836	28.9	59,386	41,440	43.3
Diluted EPS (TWD)	25.1	23.8	5.5	30.0	27.9	7.4	129.9	116.2	11.8	227.6	175.8	29.4	329.3	228.8	43.9
Gross margins (%)	57.5	60.5	(3.0)	58.0	60.2	(2.2)	58.5	61.0	(2.5)	59.9	61.8	(1.9)	60.9	62.1	(1.2)
OP margins (%)	51.5	54.5	(3.0)	52.4	54.6	(2.2)	53.1	55.6	(2.5)	55.2	57.0	(1.8)	56.4	57.5	(1.1)
Net margins (%)	42.2	43.6	(1.3)	43.0	43.8	(0.8)	43.3	44.3	(1.0)	44.7	45.5	(0.8)	45.3	45.8	(0.5)

Source: Company, Aletheia Capital, TAG

FIGURE 22: HON PRECISION - ESTIMATE VERSUS CONSENSUS

(TWD mn)	FY1Q26E			FY2Q26E			FY26E			FY27E			FY28E		
	Aletheia	Street	Var (%)	Aletheia	Street	Var (%)	Aletheia	Street	Var (%)	Aletheia	Street	Var (%)	Aletheia	Street	Var (%)
Sales	10,725	10,725	0.0	12,580	11,248	11.8	54,132	48,423	11.8	91,819	71,467	28.5	130,986	119,792	9.3
Gross profit	6,167	6,072	1.6	7,296	6,421	13.6	31,651	27,750	14.1	54,968	41,086	33.8	79,720	71,851	11.0
Operating profit	5,523	5,052	9.3	6,592	5,603	17.6	28,734	24,581	16.9	50,661	37,264	36.0	73,822	65,287	13.1
Net profit	4,529	4,219	7.4	5,405	4,564	18.4	23,434	19,984	17.3	41,039	30,287	35.5	59,386	53,073	11.9
Diluted EPS (TWD)	25.1	23.4	7.4	30.0	25.3	18.4	129.9	110.8	17.3	227.6	167.9	35.5	329.3	294.3	11.9
Gross margins (%)	57.5	56.6	0.9	58.0	57.1	0.9	58.5	57.3	1.2	59.9	57.5	2.4	60.9	60.0	0.9
OP margins (%)	51.5	47.1	4.4	52.4	49.8	2.6	53.1	50.8	2.3	55.2	52.1	3.0	56.4	54.5	1.9
Net margins (%)	42.2	39.3	2.9	43.0	40.6	2.4	43.3	41.3	2.0	44.7	42.4	2.3	45.3	44.3	1.0

Source: Company, Aletheia Capital, TAG

Valuation and recommendation

We have a high conviction Buy rating on Hon Precision and list the stock in our Alpha portfolio, supported by what we expect will be a multi-year content growth cycle for Hon's FT handler and ATC systems. Despite a planned 40% capacity expansion, we believe tight supply conditions will persist through 2028, underscoring robust demand visibility. In addition, the company is well positioned to benefit from the acceleration in the adoption of SLT solutions by ASIC customers over the next three years. This is underpinned by its long-standing relationships with leading OSATs and a strategic pivot to engage earlier with customers during the NPI stage, particularly within its SLT business.

We raise our target price for Hon Precision from NT\$6,200 to NT\$10,000. This is based on 30x SOTP PER applied to FY28E earnings; our prior TP was based on 35x SOTP PER applied to FY27E earnings. As detailed in Figure 23, we apply a 31x PER to the AI/HPC-related handler system segment, which we estimate will account for 90% of Hon's revenue mix over the next two years. This is expected to deliver 40–80% YoY revenue growth. For the non-AI/HPC segment, we apply a 20x PER, reflecting that we expect acceleration in growth in mobile handler systems, driven by rising WMCM adoption and an increase in FT test content versus wafer probing for Qualcomm and MediaTek APs.

We believe our target price reflects Hon's opportunity to capture the anticipated uptick in tester demand driven by the transition from Blackwell to Rubin architecture and expanded testing capacity for AI/HPC ASICs. The approach is also consistent with our valuation methodology for Advantest, the leader in wafer probe/FT for AI/HPC/ASIC devices, which reflects its FY28 PER earnings potential (most of Hon's machines are attached to Advantest's systems, with an attached rate of almost one to one between the companies). Like Hon Precision, Advantest is also listed in our Alpha portfolio.

FIGURE 23: HON PRECISION – SOTP VALUATION

Revenue (NT\$m)		FY25A	FY26E	FY27E	FY28E
AI/HPC		23,611	46,283	82,637	121,162
Non-AI/HPC		6,660	7,849	9,182	9,824
Total		30,271	54,132	91,819	130,986
Revenue %					
AI/HPC		78%	85%	90%	92%
Non-AI/HPC		22%	15%	10%	8%
Revenue YoY					
AI/HPC			96%	79%	47%
Non-AI/HPC			18%	17%	7%
Total			79%	70%	43%
FD-EPS (NT\$)					
AI/HPC		75.5	130.0	227.6	329.3
Non-AI/HPC		61.2	115.0	211.6	314.5
		14.4	14.9	15.9	14.8
PER Multiple					
AI/HPC testing	31x	1,886	3,546	6,525	9,696
Non-AI/HPC testing	20x	294	306	327	304
SOTP (NT\$)		2,181	3,852	6,851	10,000
Implied PER		29x	30x	30x	30x

Source: Company, Aletheia/TAG

FIGURE 24: HON PRECISION – PEER VALUATION MARIX

May 10, 2026	M. Cap	PER (X)		EPS Growth (%)		EV/EBITDA (X)		PBR (X)		ROE (%)		Div. yield (%)	
	(US\$ bn)	FY1	FY2	FY1	FY2	FY1	FY2	FY1	FY2	FY1	FY2	FY1	FY2
Hon Precision* (Dec FYE)	33.2	44.6	25.5	72.0	75.1	33.8	18.8	14.6	10.3	36.2	47.3	1.0	1.1
Chroma* (Dec FYE)	30.2	82.3	67.2	-1.5	22.5	69.5	56.2	31.0	27.2	41.4	43.1	0.6	1.0
Disco* (Mar FYE)	51.9	63.7	55.9	12.2	13.9	42.7	37.6	12.7	11.1	19.9	19.9	0.7	0.7
Shibaura* (Mar FYE)	2.2	18.0	12.6	47.2	42.4	8.8	1.2	4.7	3.7	25.9	29.2	2.2	3.2
Advantest* (Mar FYE)	139.6	35.0	27.3	78.8	28.4	25.2	19.9	18.0	11.6	65.1	51.8	0.2	0.2
Teradyne* (Dec FYE)	56.3	39.7	25.6	127.9	54.9	32.6	21.6	17.1	12.7	42.7	49.7	0.2	0.2
BESI (Dec FYE)	24.9	68.1	48.9	130.7	39.2	55.3	39.5	39.2	31.6	64.2	71.4	1.0	1.4
ASMPT (Dec FYE)	9.0	43.1	29.5	226.3	46.4	25.1	18.5	3.8	3.5	9.4	12.5	1.2	1.7
Average (excl. Hon Precision)		50.0	38.1	88.8	35.4	37.0	27.8	18.1	14.5	38.4	39.7	0.9	1.2

Source: Company, Aletheia/TAG. *Companies under Aletheia/TAG coverage

Earnings Model

FYE Dec	Mar-25	Jun-25	Sep-25	Dec-25	Mar-26	Jun-26	Sep-26	Dec-26	Mar-27	Jun-27	Sep-27	Dec-27	Dec-25	Dec-26	Dec-27	Dec-28
Sales mix, TWD mn	FY1Q25	2Q25	3Q25	4Q25	FY1Q26E	2Q26E	3Q26E	4Q26E	FY1Q27E	2Q27E	3Q27E	4Q27E	FY25A	FY26E	FY27E	FY28E
Total revenue	5,913	6,885	8,198	9,276	10,725	12,580	14,124	16,703	19,785	22,077	24,250	25,707	30,271	54,132	91,819	130,986
Total equipment shipment																
-End-of-period total capacity (unit/month)	185	200	220	245	260	280	340	380	400	440	450	470	245	380	470	510
+Blended ASP (NT\$mn)	13.4	13.5	14.2	14.7	15.4	16.6	17.4	18.0	18.3	18.9	19.2	19.3	14.0	16.9	19.0	21.3
P&L TWD mn	FY1Q25	2Q25	3Q25	4Q25	FY1Q26E	2Q26E	3Q26E	4Q26E	FY1Q27E	2Q27E	3Q27E	4Q27E	FY25A	FY26E	FY27E	FY28E
Revenue	5,913	6,885	8,198	9,276	10,725	12,580	14,124	16,703	19,785	22,077	24,250	25,707	30,271	54,132	91,819	130,986
COGS	2,419	2,847	3,468	4,423	4,558	5,283	5,791	6,848	7,993	8,875	9,700	10,283	13,157	22,481	36,851	51,267
Gross profit	3,494	4,037	4,730	4,852	6,167	7,296	8,333	9,855	11,792	13,202	14,550	15,424	17,114	31,651	54,968	79,720
OPEX	360	394	565	749	644	704	734	835	950	1,060	1,116	1,183	2,068	2,918	4,307	5,897
- R&D	208	163	239	430	279	302	311	367	396	442	461	488	1,039	1,259	1,786	2,489
- SGA	218	185	276	398	311	352	381	418	495	552	582	617	1,077	1,462	2,246	3,027
OP profit	3,133	3,643	4,166	4,104	5,523	6,592	7,599	9,020	10,842	12,142	13,435	14,242	15,046	28,734	50,661	73,822
Non-OP	102	-487	359	565	181	215	190	194	296	273	225	232	538	780	1,026	971
- net interest	65	56	59	127	127	127	127	127	127	127	127	127	307	510	510	510
- others	37	-543	300	438	54	88	62	66	168	146	97	105	231	270	516	461
PTI	3,235	3,156	4,524	4,669	5,705	6,807	7,788	9,214	11,138	12,416	13,660	14,474	15,584	29,514	51,687	74,793
Tax and hybrid capital	-667	-690	-907	-958	-1,175	-1,402	-1,604	-1,898	-2,294	-2,558	-2,814	-2,982	-3,222	-6,080	-10,647	-15,407
Net profit	2,568	2,466	3,617	3,711	4,529	5,405	6,184	7,316	8,843	9,858	10,846	11,492	12,362	23,434	41,039	59,386
FD EPS, Diluted (NT\$)	15.8	15.2	22.3	22.1	25.1	30.0	34.3	40.6	49.0	54.7	60.1	63.7	75.5	129.9	227.6	329.31
Profitability	FY1Q25	2Q25	3Q25	4Q25	FY1Q26E	2Q26E	3Q26E	4Q26E	FY1Q27E	2Q27E	3Q27E	4Q27E	FY25A	FY26E	FY27E	FY28E
Gross margin	59.1%	58.6%	57.7%	52.3%	57.5%	58.0%	59.0%	59.0%	59.6%	59.8%	60.0%	60.0%	56.5%	58.5%	59.9%	60.9%
OPN margin	53.0%	52.9%	50.8%	44.2%	51.5%	52.4%	53.8%	54.0%	54.8%	55.0%	55.4%	55.4%	49.7%	53.1%	55.2%	56.4%
PTI margin	54.7%	45.8%	55.2%	50.3%	53.2%	54.1%	55.1%	55.2%	56.3%	56.2%	56.3%	56.3%	51.5%	54.5%	56.3%	57.1%
Net margin	43.4%	35.8%	44.1%	40.0%	42.2%	43.0%	43.8%	43.8%	44.7%	44.7%	44.7%	44.7%	40.8%	43.3%	44.7%	45.3%
YoY chg.	FY1Q25	2Q25	3Q25	4Q25	FY1Q26E	2Q26E	3Q26E	4Q26E	FY1Q27E	2Q27E	3Q27E	4Q27E	FY25A	FY26E	FY27E	FY28E
Revenue	142.4%	128.7%	128.9%	87.0%	81.4%	82.7%	72.3%	80.1%	84.5%	75.5%	71.7%	53.9%	116.3%	78.8%	69.6%	42.7%
Gross profit	188.1%	120.5%	139.8%	80.8%	76.5%	80.7%	76.2%	103.1%	91.2%	80.9%	74.6%	56.5%	122.3%	84.9%	73.7%	45.0%
OPN profit	220.3%	143.0%	158.1%	87.1%	76.3%	80.9%	82.4%	119.8%	96.3%	84.2%	76.8%	57.9%	139.4%	91.0%	76.3%	45.7%
PTI	183.3%	102.7%	201.9%	87.8%	76.3%	115.7%	72.1%	97.3%	95.2%	82.4%	75.4%	57.1%	133.2%	89.4%	75.1%	44.7%
Net profit	179.3%	101.9%	206.6%	88.8%	76.4%	119.2%	71.0%	97.1%	95.2%	82.4%	75.4%	57.1%	133.9%	89.6%	75.1%	44.7%
FD EPS	181.9%	104.8%	209.0%	77.8%	58.6%	96.7%	53.5%	83.4%	95.2%	82.4%	75.4%	57.1%	130.8%	72.0%	75.1%	44.7%

Source: Company, Aletheia/TAG

Financial statements

FYE Dec (TWD mn)	FY23A	FY24A	FY25A	FY26E	FY27E	FY28E
Revenue	9,489	13,992	30,271	54,132	91,819	130,986
COGS	-4,823	-6,293	-13,157	-22,481	-36,851	-51,267
Gross profit	4,667	7,700	17,114	31,651	54,968	79,720
Operating expense	-901	-1,415	-2,068	-2,918	-4,307	-5,897
EBITDA	3,853	6,568	15,346	29,127	51,385	74,581
Operating profit	3,766	6,285	15,046	28,734	50,661	73,822
Interest income (expense)	68	160	307	510	510	510
Other income (expense)	52	238	231	270	516	461
Profit before tax	3,887	6,683	15,584	29,514	51,687	74,793
Tax	-819	-1,397	-3,222	-6,080	-10,647	-15,407
Net profit	3,068	5,286	12,362	23,434	41,039	59,386
FD EPS TWD	18.8	32.7	75.5	129.9	227.6	329.3

Change YoY

Revenue	-30%	47%	116%	79%	70%	43%
Gross profit	-25%	65%	122%	85%	74%	45%
Operating profit	-25%	67%	139%	91%	76%	46%
Net profit	-30%	72%	134%	90%	75%	45%
EPS	-31%	74%	131%	72%	75%	45%

CASH FLOW

FYE Dec (TWD mn)	FY23A	FY24A	FY25A	FY26E	FY27E	FY28E
Net profit	3,068	5,286	12,362	23,434	41,039	59,386
Dep. & amortization	35	46	69	123	209	298
Chg in WC	2,297	-2,896	2,671	-5,645	-12,027	-12,473
Others	-116	-127	1,085	0	0	0
OPN cash flow	5,283	2,308	16,186	17,912	29,221	47,210
Capex	-230	-133	-286	-562	-971	-1,393
Chg in investment	0	0	0	0	0	0
Others	-676	-1,189	854	0	0	0
Investing CF	-906	-1,322	569	-562	-971	-1,393
Chg in debt	-9	147	-159	0	0	0
Equity raised	0	0	-21	0	0	0
Dividend paid	-2,398	-4,800	-3,636	-9,900	-10,796	-11,695
Others	0	894	34,466	0	0	0
Financing CF	-2,408	-3,759	30,650	-9,900	-10,796	-11,695
FX & other adj	-2	4	-2	0	0	0
Chg in cash	1,968	-2,769	47,402	7,450	17,454	34,122
Beginning cash	6,825	8,794	6,025	53,427	60,877	78,331
End cash	8,794	6,025	53,427	60,877	78,331	112,453
FCF	5,591	2,888	16,105	19,662	32,452	51,977

Source: Company, Aletheia/TAG

FYE Dec (TWD mn)	FY23A	FY24A	FY25A	FY26E	FY27E	FY28E
Cash/ST investment	9,498	8,045	54,228	61,679	79,132	113,254
Inventory	4,092	8,444	12,641	18,307	30,506	43,155
Receivable	1,111	2,913	2,935	5,279	9,043	13,029
Current assets	14,744	19,537	70,195	85,657	119,073	169,829
Fixed assets	1,778	1,898	2,163	2,504	3,098	3,950
Goodwill & intangibles	0	0	0	98	267	510
Other LT assets	355	431	912	912	912	912
LT assets	2,133	2,329	3,075	3,514	4,277	5,372
Total assets	16,877	21,866	73,270	89,170	123,350	175,201

Payable	536	3,181	3,402	5,768	9,704	13,865
Other liability	3,989	4,847	11,626	11,626	11,626	11,626
Current liabilities	4,525	8,028	15,028	17,394	21,330	25,491
LT liabilities	37	106	211	211	211	211
Total liabilities	4,562	8,134	15,239	17,605	21,541	25,702
Common shares	1,600	1,616	1,799	1,799	1,799	1,799
Retained earnings	10,696	11,182	19,908	33,442	63,686	111,376
Equity	12,315	13,732	58,031	71,565	101,809	149,499
Liability&Equity	16,877	21,866	73,270	89,170	123,350	175,201

RATIO & PER SHARE DATA

FYE Dec (TWD mn)	FY23A	FY24A	FY25A	FY26E	FY27E	FY28E
Gross margin (%)	49.2	55.0	56.5	58.5	59.9	60.9
EBITDA margin (%)	40.6	46.9	50.7	53.8	56.0	56.9
Op. Profit margin (%)	39.7	44.9	49.7	53.1	55.2	56.4
Net Profit margin (%)	32.3	37.8	40.8	43.3	44.7	45.3
ROE (%)	25.6	40.6	34.5	36.2	47.3	47.3
ROA (%)	18.4	27.3	26.0	28.9	38.6	39.8
Gross gearing (%)	27.0	37.2	20.8	19.7	17.5	14.7
Net gearing (%)	-71.4	-42.7	-92.0	-85.0	-76.9	-75.2
Asset turn (x)	0.6	0.7	0.6	0.7	0.9	0.9
Leverage (x)	1.4	1.5	1.3	1.3	1.2	1.2
Payable days	57	108	91	94	96	99
Receivable days	70	52	35	36	36	36
Inventory days	350	364	292	297	302	307
EPS, diluted	18.8	32.7	75.5	129.9	227.6	329.3
BVPS	75.5	85.0	354.6	396.8	564.6	829.0
Net cash per share	53.9	36.3	326.2	337.3	434.1	623.3
SPS	58.2	86.6	185.0	300.2	509.2	726.3
FCFPS	34.3	17.9	98.4	109.0	180.0	288.2
DPS	30.0	22.5	55.0	60.0	65.0	70.0

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